

Executive Summary: Teen Phone Addiction Analysis

1. Objective

This comprehensive report outlines the development, evaluation, and key insights derived from machine learning models designed to predict phone addiction levels among teenagers. The primary goal is to isolate the critical behavioral, demographic, and psychological drivers of phone addiction to provide actionable, data-driven insights for parents, educators, and mental health professionals.

2. Methodology & Model Evaluation

We systematically analyzed a rich dataset encompassing teenage demographic profiles, digital usage patterns, academic performance metrics, and mental health indicators. Several regression algorithms were evaluated to predict the continuous 'Addiction_Level' variable, including Linear Regression, Support Vector Machines, Decision Trees, Random Forest, XGBoost, and CatBoost.

The CatBoost Regressor emerged as the definitive best-in-class model, achieving an outstanding R-squared score of 0.98 and a Cross-Validation Accuracy of nearly 98%. It exhibited the lowest Mean Absolute Error (MAE: 0.18) and Root Mean Squared Error (RMSE: 0.24), significantly outperforming baseline models and establishing itself as a highly reliable predictive tool.

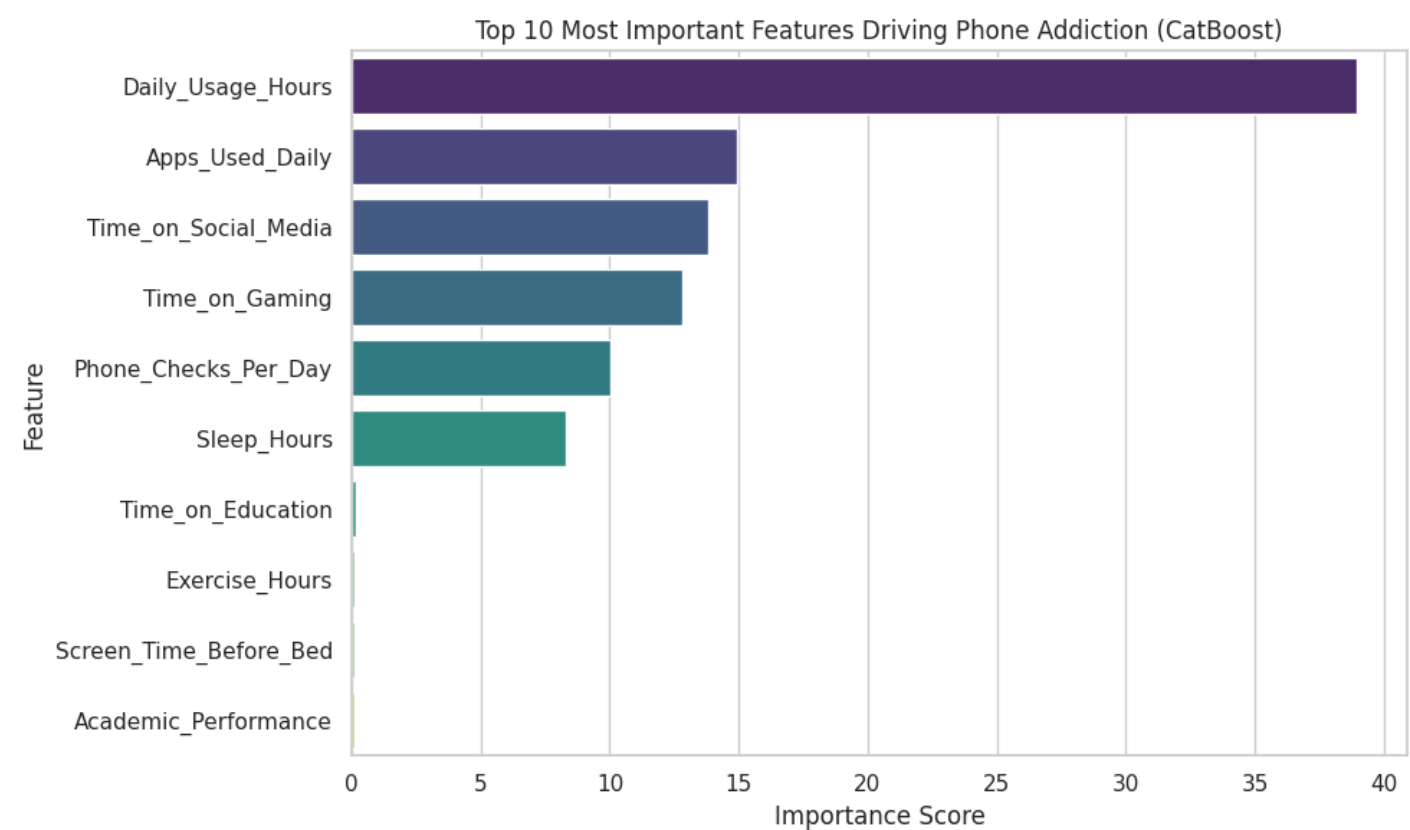
3. Critical Drivers of Addiction (Feature Importance)

By extracting the feature importance scores from the CatBoost model, we identified the most critical factors influencing addiction levels:

- Daily Usage Hours (38.9%): The overwhelming primary driver. Absolute screen time is directly correlated with addiction severity.
- Apps Used Daily (15.0%): High engagement with a wide variety of applications indicates fragmented attention and deeper integration into the digital ecosystem.
- Time on Social Media (13.8%) & Gaming (12.8%): Specific content consumption, particularly highly stimulating interactive media, strongly drives addictive behavior.
- Phone Checks Per Day (10.0%): The habitual, compulsive nature of checking the device serves as a major psychological indicator.
- Sleep Hours (8.3%): A critical lifestyle factor. Reduced sleep is intimately linked with higher addiction levels, creating a negative feedback loop.

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4. Visual Summary of Feature Importance



5. Strategic Recommendations

- Intervention Focus: Strategies must prioritize reducing 'Daily Usage Hours' and the breadth of 'Apps Used Daily'. Limiting raw exposure is the most effective immediate step.
- Habit Breaking: Interventions should actively disrupt the habitual 'Phone Checks Per Day'. Setting designated 'phone-free' intervals can help break this compulsive cycle.
- Holistic Approach: Programs must extend beyond the device to address lifestyle factors. Ensuring adequate 'Sleep Hours' and monitoring for related signs of 'Depression' or 'Anxiety' are essential for comprehensive treatment.
- Educational Support: Schools and communities can use these metrics to identify at-risk behavior patterns early and provide targeted digital literacy programs.

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6. Project Assessment and Real-World Applications

Project Assessment:

This project successfully demonstrates a comprehensive workflow for building and evaluating regression models. Key strengths include clear problem definition, robust data preprocessing (handling irrelevant columns and encoding), exploration of multiple algorithms (Linear to CatBoost), and rigorous evaluation (R2, RMSE, K-Fold CV). The feature importance analysis provides actionable insights, supported by clear visualizations.

Real-World Application:

- Parents: Monitor and manage Daily Usage Hours and specific activities (Social Media, Gaming). Recognize the impact of Sleep Hours over simply limiting Parental Control.
- Educators: Implement educational programs focusing on healthy habits and risky usage patterns. Support students showing links between digital habits and Academic Performance.
- Healthcare Professionals: Assess addiction risk based on identified key features (Usage Hours, Checks Per Day). Address interconnected factors like Sleep and Depression Levels as part of comprehensive interventions.